

Main effects and interactions

- *Sensible* statistical modeling requires that interaction terms are added to the linear predictor only when the individual terms, known as **main effects**, have also been added
- In our example we would then have an intercept, an age effect, a state effect, and their interaction
 - The interaction comes to explain what is beyond an age and state effects, some crossing(s) of age and state that behave differently relative to what the main effects would suggest

- In R for variables say `x1` and `x2` the **model formula**:
 - `x1*x2` would add in the linear predictor `x1`, `x2` and their interaction
 - `x1:x2` denotes just the interaction

- The previous examples involve **two-way interactions**
- In the same vain we can define **multi-way interactions**
 - Operationally this is again nothing but constructing a new features as the, so-called, **tensor product** (the extension of outer in dim larger than 2) of the tree constituent covariates
 - But! again with particular type of dependence with the constituent main effects and two-way interactions

- In the terminology of ¹ such multi-way interactions are called **deep interactions**
- Again, it is appropriate to add deep interactions once main effects and more shallow, so to speak, interactions have been included
 - Hence, we would add a three way interaction among age, state and ethnicity if main effects (age, state, ethnicity) and two-way interactions (age-state, age-ethnicity, state-ethnicity) have already been included

- When we add as main effects a numerical, say x_1 , and a categorical, say x_2 , covariates, and we also include their interaction, then identifiability issues arise that involve the varying slopes
- If say β_1 is the slope for x_1 and $\beta_{2,g}$ is the varying slope when the observation i takes level g in covariate x_2 , the contribution to the linear predictor in terms of x_1 is

$$(\beta_1 + \beta_{2,g})x_{1,i}$$

Interactions and nesting

- We say that one variable is nested within another when, by design, we can only observe certain cells in the contingency table defined by their outer product
 - E.g, if one variable is `student_id` and the other `school_id` then because each student belongs to a single school, we can only have observations in some of the available cells
 - E.g., the experimental design in Section 2.1 in ¹

- In particular, if we were to organize the data in a matrix, and have the columns organized by `school_id` and organize rows so that students from the same school are placed in consecutive rows following the same order as the columns, and write 1 for every cells that it is observed in the dataset and 0 otherwise, then the data matrix would be **block-diagonal**
- More generally, we can define nested variables as those for which the associated data occupancy matrix can be row/column-permuted to block-diagonal
- Notice that by construction, interaction effects define variables that are nested within the constituent main effects

Identifiability, pt 2

- For each main categorical effect there will be 1 parameter too many
- If there are nested covariates there are more redundant parameters
 - In the example with schools and students, we could add a constant to the school parameter and remove it from all student parameters from that school and keep the linear predictor unchanged
 - Effectively, there is no way to tell, say, if the school is very good or all its students are very good

- In this example we could either drop the `school_id` from the linear predictor or add both `student_id` and `school_id` but add identifiability constraints
 - What would make more sense here is to constrain the coefficients of the nested variable, `student_id`, to sum to zero

- Of course interactions by construction create nested covariates
 - Consider our example of `age:state` which is nested within both `age` and `state`
- It is (sort of) easy to see that if we have categorical covariates `x1`, with G_1 levels, and `x2`, with G_2 levels, the model `x1*x2` would need
 - 1 constraint for `x1`
 - 1 constraint for `x2`
 - $G_1 + G_2 - 1$ for `x1:x2`

- Default in **R** is to use the “first” level of each main effect as pivot, hence set the corresponding parameters to 0, and then set to 0 also the parameters that correspond to any interactions that involve the pivot levels
- In an analogous manner are dealt with interactions between a numerical and categorical covariate

- A summary of the previous is:
 - Identifiability constraints can be imposed in different ways
 - They are not relevant only when dealing with interactions but also with nested covariates (interactions is a special case of that)
 - A different perspective is that the resulting design matrix is **rank deficient**, a situation known as **collinearity**
 - Notice that, for example, including state-level contextual covariates (e.g, `\texttt{z_trnprv}`) and state indicators (`stt`) also creates (near) collinearity

- For more details of how this is represented mathematically and implemented in **R** see Sections 1.6, 1.7, 1.8 in ¹, but mainly Section 6.2 in ²
- Identifiability constraints - but more broadly the implications of unidentifiability - will be a returning topic in these lectures
- In the sequel we will adopt a **probabilistic inference** approach, using **random effect modeling**, which to some extent (or better said, initially) will remove such problems

An illustration

- For the sake of an example here is a *very* simple model on Application 1

```
glm(formula = vote ~ scale(z_incstt) +  
as.factor(age)*z_trnprv, family = binomial, data  
= gg_dat)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.087650	0.016118	5.438	5.39e-08	***
scale(z_incstt)	-0.006383	0.008594	-0.743	0.4576	
as.factor(age)2	0.613787	0.021467	28.592	< 2e-16	***
as.factor(age)3	1.019630	0.021293	47.886	< 2e-16	***
as.factor(age)4	1.066750	0.026286	40.583	< 2e-16	***
z_trnprv	0.361567	0.035099	10.301	< 2e-16	***
as.factor(age)2:z_trnprv	0.091889	0.045941	2.000	0.0455	*
as.factor(age)3:z_trnprv	0.110529	0.045562	2.426	0.0153	*
as.factor(age)4:z_trnprv	0.117830	0.056919	2.070	0.0384	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 98696 on 79147 degrees of freedom
Residual deviance: 95265 on 79139 degrees of freedom
AIC: 95283

Number of Fisher Scoring iterations: 4

- This is another very simple model:

```
glm(formula = vote ~
as.factor(stt)*as.factor(age), data=gg_dat, family=binom)
```

Coefficients:

	Estimate	Std. Error	z value				
(Intercept)	0.1238695	0.1438671	0.861	as.factor(stt)51	-0.0438268	0.1916335	-0.229
as.factor(stt)2	0.4373502	0.1958161	2.233	as.factor(age)2	0.5179844	0.1896336	2.732
as.factor(stt)3	-0.1592714	0.1959653	-0.813	as.factor(age)3	1.0003384	0.1881992	5.315
as.factor(stt)4	-0.4714119	0.2070243	-2.277	as.factor(age)4	0.9465719	0.2206216	4.290
as.factor(stt)5	-0.1935301	0.1567511	-1.235	as.factor(stt)2:as.factor(age)2	-0.4306890	0.2563064	-1.680
as.factor(stt)6	-0.0803844	0.1776864	-0.452	as.factor(stt)3:as.factor(age)2	-0.0713867	0.2607691	-0.274
				as.factor(stt)4:as.factor(age)2	0.0701485	0.2690939	0.261

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 98696 on 79147 degrees of freedom
Residual deviance: 94606 on 78944 degrees of freedom
AIC: 95014

Number of Fisher Scoring iterations: 4

GLMMs

From varying to random coefficients

- We move to a framework that has revolutionized applied statistics and it is used by all areas of science - including (and quite extensively) by tech industry
- It will provide a common framework to address a number of issues associated with GLMs - especially when dealing with **high-dimensional categorical covariates** and deep interactions
- It will also provide a continuum between linear and bilinear terms

- The first thing to note is that varying coefficient models impose a particular **grouping** on the coefficients
 - E.g., the age main effects, or the set of age-turnout interactions in Application 1
 - It might also make sense to consider set of pairs of varying intercept (for age main effect) and varying slope (by its interaction with turnout) as a group of 4 bivariate coefficients
- Collectively, each such group defines a **source of variation**

- The approach we will adopt is to model varying coefficients as random variables, and we will call them **random coefficients**
- The *default* approach will be to model the coefficients associated to a given group as **exchangeable Gaussian random variables** with mean 0 and a given variance (in the case of a single coefficient) or covariance matrix (in the case of multivariate coefficients)
 - Such variance terms are **hyperparameters** and will need to be learned from data

- We will switch to denoting by α , when scalar, or $\boldsymbol{\alpha}$, when multivariate, such random coefficient(s), as opposed to β that will be used for the non-random coefficients
 - These are typically called **fixed effects** and we will do the same here
- In principle, fixed effects can also include varying coefficients, although our *default* approach will be to model *all* varying coefficients as random
 - We will discuss in the sequel the issue of fixed vs random

Multilevel notation for random coefficients

- It will be helpful to introduce an intuitive notation that can be used to describe models with different categorical covariates hence various sets of random coefficients
- One which is very popular in different application areas, e.g. political science, is the so-called **multilevel notation**, used extensive e.g. in ¹

- It is arguably the easiest one to communicate to non-experts, especially for relatively simple models where the categorical covariates are referred to by what they represent in “real life”
- In this notation we can avoid the use of one-hot-encoding binary features to represent categorical covariates
 - Of course, these are important for computational implementations but less appropriate to communicate and understand model structures and assumptions

- According to this, we write, e.g., α_j^{stt} to refer to the random intercept associated with level j of the categorical covariate “state”
 - Hence, random intercepts are modeled as:

$$\alpha_j^{\text{stt}} \sim \mathcal{N}(0, \sigma_{\text{stt}}^2), \quad j = 1, \dots, 51$$

- Additionally, for a categorical covariate with levels $j = 1, \dots, G$, and an observation i , $j[i]$ denotes the value i has on the covariate

- The random intercept associated with a state-age interaction, for state level j and age level h , is denoted by $\alpha_{j,h}^{\text{stt,age}}$
- Therefore, as an example, the linear predictor for a model with global intercept, age and state main effects and their interaction is:

$$\eta_i = \beta_0 + \alpha_{j[i]}^{\text{stt}} + \alpha_{h[i]}^{\text{age}} + \alpha_{j[i],h[i]}^{\text{stt,age}}$$

Random vs fixed effects

- What to treat as random as opposed to fixed effect is a common source of confusion and depending on which area of science one comes from, or even which sub-field, this question might appear: fundamental/immaterial, obvious/incomprehensible, foundational/computational!

- It is not our priority to discuss this point thoroughly
 - Rather, we will list in the sequel a summary of some of the advantages of random effect modeling
 - Still, we will provide some high-level discussion of different perspectives and pointers to the literature to help those interested understand the nature of the question and some of its answers
- The article ¹ and the contributed discussion therein is a good input for reflection on this topic, and we align some of our choices with Gelman's approach

- From a **Bayesian inference** standpoint, all unknown parameters in the model are treated as random variables and are modeled using distributions, known in this context as **prior distributions**
- According to this perspective, it is obvious that distributions are specified for varying coefficients, but also for the fixed ones

- External information can be used to inform the specification of said priors, such as the fact that certain subsets of parameters form groups
- Additionally, in absence of information to differentiate apriori different parameters, **exchangeability** is sensible and can also be justified theoretically (see, e.g., Chapter 4 in ¹)

- Operationally, the approach we adopt is to specify prior distributions for all unknowns, including varying coefficients, fixed effects and hyperparameters
 - For fixed effects we use so-called flat priors, therefore the distinction between random and fixed effects is one of the type of priors used (more and less informative respectively)
 - We discuss priors for hyperparameters later on

- This is in the spirit of ¹: “Our approach is to use variance components modeling for all rows of the table, even for those sources of variation that have commonly been regarded as fixed effects.”
- However, our variational methods adapt fairly immediately when only varying coefficients are modeled probabilistically and fixed effects and hyperparameters are treated as non-random

- On the other hand, it is important to understand that random effect modeling is **not** the same as Bayesian modeling - despite the similarities
 - Hence, it is widely adopted by communities that have very little sympathy towards Bayesian inference
- In particular, due to the inherent replication in a varying coefficient setting, certain assumptions about the random effects can be tested from the observed data
 - Consider the example of students and schools, and large student samples per school